

Hybrid Quantum-Classical Framework for Multi-Label Bengali Emotion Classification

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Abstract

Multi-label emotion classification of code-mixed and low-resource languages such as Bengali remains a challenging task because of limited labelled corpora, severe class imbalance, and the need to recognise multiple co-occurring emotions per comment. This paper investigates hybrid quantum-classical models that combine pretrained transformer representations with trainable variational quantum circuits (VQCs) for six-way multi-label emotion classification on the EmoNoBa dataset. We design two custom four-qubit VQC architectures—a cross-coupling entangling scheme (Custom 1) and a re-uploading encoding scheme with a rotating ansatz (Custom 2)—and evaluate them with two Bengali encoders (IndicBERTv2 and BanglaBERT) across 2, 4, and 8-qubit registers. In addition to the per-label macro, micro, and weighted F1-scores, we report the Exact Match Ratio (EMR) and an instance-level partial-match F1-score. The quantum variants substantially improve the weaker BanglaBERT baseline, raising its macro F1-score from 0.2101 to 0.4850 and its EMR from 0.0000 to 0.4630, while the best overall macro F1-score of 0.5165 and the highest EMR of 0.4577 are achieved with IndicBERTv2 and Custom 2 at eight qubits. Our results indicate that the benefit of additional qubits depends critically on the compatibility of the encoding, entanglement pattern, and classical representation.

Keywords: hybrid quantum-classical machine learning, variational quantum circuits, multi-label emotion classification, Bengali natural language processing, low-resource languages, exact match ratio

1 Introduction

Multi-label emotion classification is an important natural language processing (NLP) task with applications in sentiment-aware dialogue systems, social-media analysis, public-opinion monitoring, and human-computer interaction. Unlike binary sentiment analysis, which assigns a single polarity, multi-label emotion recognition requires a model to predict a set of emotions for each input, since closely related emotions frequently co-occur. The problem becomes substantially more difficult for low-resource languages such as Bengali, because labelled corpora, language-specific pretrained models, balanced emotion datasets, and standardized benchmarks are comparatively limited [1, 2]. Minority emotion classes are easily suppressed by dominant labels, and the informal, noisy style of social-media text compounds the difficulty.

In parallel, near-term quantum machine learning (QML) has emerged as a promising paradigm in which parameterized quantum circuits (PQCs) map classical feature vectors into high-dimensional quantum Hilbert spaces [8]. Variational quantum circuits—PQCs with trainable rotation parameters updated by gradient descent—have been applied to supervised learning [5], and the data re-uploading paradigm interleaves classical feature encoding layers $U(x)$ with trainable

unitaries $U(\theta)$ to increase the expressivity of low-qubit circuits [4]. Recent analyses of parameterized circuit expressibility and entanglement [6] and of hybrid quantum-classical training dynamics [9] provide a theoretical basis for designing trainable quantum feature extractors. These developments motivate the question of whether a variational quantum circuit, inserted as a trainable bottleneck inside a classical transformer architecture, can improve multi-label emotion classification for Bengali.

In this paper, we present a hybrid quantum-classical framework that combines a pretrained Bengali transformer encoder with a trainable variational quantum circuit for multi-label emotion classification on the EmoNoBa dataset [2]. The contributions of this paper are as follows:

- We develop a hybrid quantum-classical framework that couples a pretrained Bengali transformer encoder (IndicBERTv2 [3] or BanglaBERT [1]) with a trainable variational quantum circuit for six-way multi-label emotion classification, and a classical baseline without a quantum branch.
- We design and compare two custom four-qubit VQC architectures—a cross-coupling entangling scheme (Custom 1) and a re-uploading encoding scheme with a rotating ansatz (Custom 2)—under an identical outer framework, and evaluate both across 2, 4, and 8-qubit registers.
- We evaluate all configurations with a unified protocol covering per-label macro, micro, and weighted F1-scores, the Hamming loss, the Exact Match Ratio (EMR), and an instance-level partial-match F1-score, and analyse how encoding, entanglement, and circuit depth interact with the quality of the classical representation.

The remainder of this paper is organized as follows. Section 2 reviews related work on low-resource Bengali NLP and on quantum machine learning. Section 3 describes the EmoNoBa dataset and its label distribution. Section 4 presents the task definition, the overall architecture, the two VQC designs, the training setup, and the evaluation metrics. Section 5 describes the experimental setup and reports the results. Section 6 discusses the findings, Section 7 concludes the paper, and Section 8 outlines the limitations of the present work.

2 Related Work

Recent advances in low-resource NLP demonstrate that pretrained language models such as BanglaBERT [1] and IndicBERTv2 [3] provide strong contextual text representations for Indic languages, including Bengali. However, multi-label emotion classification remains constrained by rigid output thresholds and can suffer substantial performance degradation under strict evaluation metrics such as the Exact Match Ratio (EMR).

In the quantum domain, variational quantum circuits have been established as trainable models for supervised learning [5, 9]. Pérez-Salinas et al. introduced the data re-uploading paradigm, in which classical feature-encoding layers $U(x)$ are interleaved with trainable unitary transformations $U(\theta)$ across L layers [4]:

$$U(\theta, x) = \prod_{l=1}^L U(\theta_l) U(x). \quad (1)$$

Repeatedly injecting the input allows constrained circuits to represent more complex decision functions than a single encoding layer. Sim et al. analysed the expressibility and entangling capability of parameterized circuits and showed that the choice of gate family and entanglement structure directly affects the representational power of the ansatz [6]. Qin reviewed ansatz-design techniques for variational quantum algorithms, summarizing trade-offs between circuit depth, trainability, and hardware efficiency [10]. Beyond supervised classification, variational

circuits have been applied successfully to reinforcement learning [11] and to hardware-efficient quantum eigensolvers [7], demonstrating the broad utility of the variational ansatz. Building on these foundations, the present work combines a transformer feature extractor with a data-re-uploading VQC ansatz and per-label decision-threshold calibration, targeting class imbalance and joint subset-prediction accuracy in multi-label Bengali emotion classification.

3 Dataset

This study uses the EmoNoBa dataset, a manually annotated collection of 22,698 public Bangla comments gathered from social media platforms [2]. The comments span 12 domains, including personal matters, politics, and health, and are labelled with six fine-grained emotion categories based on the Juntto Emotion Wheel. The dataset preserves the informal and noisy characteristics of online Bangla text, making it suitable for evaluating emotion-classification models under realistic conditions.

The dataset is split into training, validation, and test partitions of 18,420, 2,047, and 2,272 comments, respectively. Table 1 reports the per-emotion label counts in the training partition. The distribution is strongly imbalanced: *Joy* (8,314) dominates, whereas *Surprise* (848) and especially *Fear* (279) are severely underrepresented. As discussed in Section 4.4, this imbalance motivates the use of per-label positive weights in the binary cross-entropy loss to prevent the minority emotion classes from being suppressed.

Table 1: Distribution of emotion labels in the training partition of the EmoNoBa dataset.

Emotion	Count	Percentage
Joy	8,314	45.14
Sadness	4,569	24.80
Love	3,786	20.55
Anger	3,542	19.23
Surprise	848	4.60
Fear	279	1.51
Total	18,420	100.00

4 Methodology

4.1 Task Definition

Given a Bengali comment x consisting of a sequence of tokens, the objective is to predict a binary label vector $y \in \{0, 1\}^K$ over $K = 6$ emotion categories—*Love*, *Joy*, *Surprise*, *Anger*, *Sadness*, *Fear*—where $y_k = 1$ indicates that emotion k is expressed in the comment and multiple emotions may co-occur. Following the convention of multi-label classification, the model produces an independent logit z_k for each emotion, which is converted to a probability via the sigmoid function $\sigma(z_k) = 1/(1 + e^{-z_k})$ and thresholded at 0.5 to obtain the predicted label $\hat{y}_k$. The goal is to maximise joint subset-prediction accuracy, measured by the Exact Match Ratio, while retaining competitive per-label performance.

4.2 Framework Description

Figure 1 provides an overview of the proposed hybrid classical-quantum framework. For both the classical baseline and the quantum variants, each comment is first tokenized with a maximum sequence length of 128 and passed through a pretrained Bengali transformer encoder,

either IndicBERTv2 or BanglaBERT, from which the 768-dimensional [CLS] representation is extracted. A trainable linear layer projects this representation to a 128-dimensional feature vector, which is then squashed by a tanh activation.

In the quantum variant, the compressed features are further projected by a linear layer into a low-dimensional vector matching the qubit register (e.g. four features for a four-qubit circuit) and fed into the variational quantum circuit described in Section 4.3. The expectation values produced by the circuit are mapped back through a post-quantum linear layer to the 128-dimensional working space, followed by a ReLU activation. In the classical baseline, the quantum block is removed and the 128-dimensional features are passed directly to the classifier. In both cases, a final linear layer maps the representation to the $K = 6$ logits, which are interpreted with sigmoid outputs for multi-label prediction.

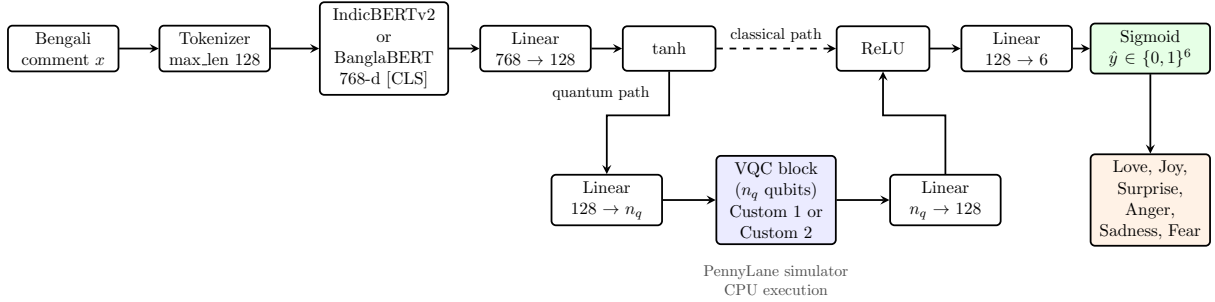


Figure 1: End-to-end framework of the hybrid classical-quantum model for multi-label Bengali emotion classification. The classical path (dashed) bypasses the quantum block and is used as the baseline; the quantum path (solid) inserts the variational quantum circuit between the compressed text representation and the post-quantum fusion layer.

4.3 VQC Designs

Following the transformer encoding, the 768-dimensional CLS embedding is projected by a trainable linear layer to a low-dimensional feature vector. Depending on the variant, the vector is squashed by tanh and fed directly into a four-qubit variational quantum circuit (VQC); the circuit outputs are mapped back through a post-quantum linear layer and a ReLU activation before the final classifier. Two custom VQCs are investigated.

Custom 1 cross-coupling VQC. This circuit acts on $n = 4$ qubits and consumes an $\mathbb{R}^4$ vector.

- (i) **Encoding.** Each feature is directly injected as a single-axis rotation $R_Y(x_i)$ on qubit i (angle embedding).
- (ii) **Ansatz.** Two variational layers are used, each consisting of a general Rot applied to every qubit followed by a controlled-NOT entangling stage. The first stage uses a nearest-neighbour ring, with CNOT gates on $(0, 1)$, $(1, 2)$, $(2, 3)$, and $(3, 0)$; the second uses a cross-coupling pattern, with CNOT gates on $(0, 2)$, $(1, 3)$, $(2, 1)$, and $(3, 0)$. The same 12-parameter weight tensor is shared by both Rot layers. No data re-uploading is performed.
- (iii) **Measurement.** The Pauli-Z expectation is measured on every qubit, producing four outputs for the post-quantum linear layer.

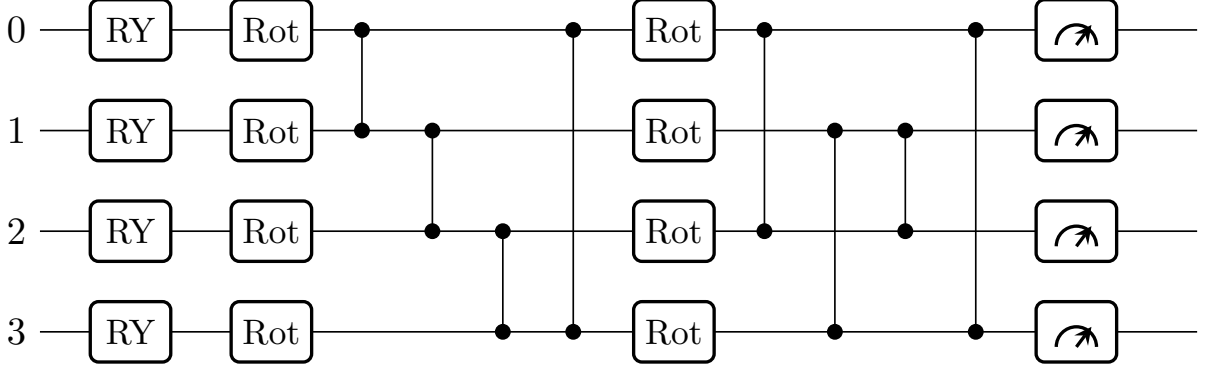


Figure 2: Four-qubit circuit architecture used for the Custom 1 VQC.

Custom 2 re-uploading VQC. This circuit acts on $n = 4$ qubits and consumes an $\mathbb{R}^8$ vector, i.e. two features per qubit.

- (i) **Encoding.** For each qubit q , its feature pair (x_{2q}, x_{2q+1}) is first rotated by a learnable phase ϕ_q :

$$x'_{2q} = s_q(\cos \phi_q x_{2q} - \sin \phi_q x_{2q+1}), \quad (2)$$

$$x'_{2q+1} = s_q(\sin \phi_q x_{2q} + \cos \phi_q x_{2q+1}), \quad (3)$$

where s_q is a second learnable amplitude per qubit. The rotated values are applied as rotation angles $R_X(x'_{2q})$ and $R_Y(x'_{2q+1})$ on qubit q .

- (ii) **Ansatz.** Each variational block applies a general single-qubit rotation $R_Z R_Y R_Z \equiv \text{Rot}$ on every qubit, followed by a ring of controlled- Z entangling gates $CZ[q, (q+1) \bmod 4]$.
- (iii) **Data re-uploading.** The data-dependent encoding is inserted before both variational blocks, i.e. encode $\rightarrow$ ansatz $\rightarrow$ encode $\rightarrow$ ansatz, so the input is re-supplied between trainable layers.
- (iv) **Measurement.** Both $\langle Z_q \rangle$ and $\langle X_q \rangle$ are measured on all four qubits, yielding $2n = 8$ expectation values that feed the post-quantum layer. The circuit contains 20 trainable parameters: 4 phase parameters, 4 scale parameters, and 12 ansatz angles.

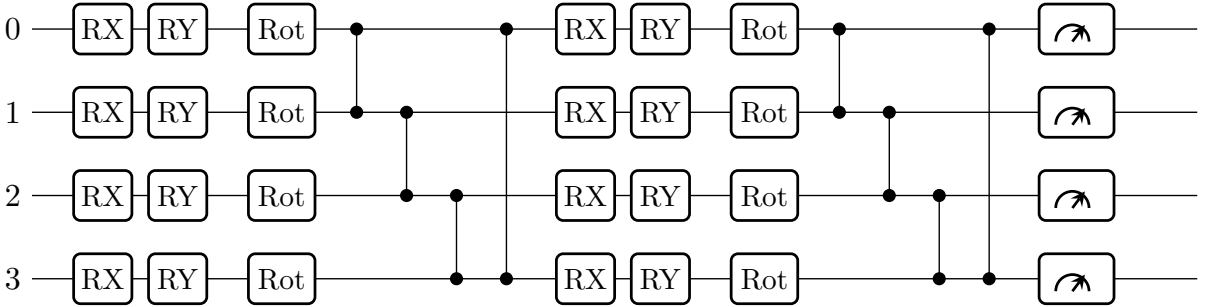


Figure 3: Four-qubit circuit architecture used for the Custom 2 VQC.

The two designs therefore contrast a cross-coupling entangling scheme without feature re-uploading (Custom 1) against a re-uploaded encoding-plus-entangling scheme (Custom 2), while keeping the outer hybrid architecture identical.

4.4 Training

All models were trained for a maximum of 25 epochs with a batch size of 8 and a maximum sequence length of 128. The AdamW optimizer was used with a learning rate of 3×10^{-5} and a weight decay of 0.01. Because the EmoNoBa emotion labels are strongly imbalanced (Table 1), the binary cross-entropy loss was weighted with per-label positive weights computed from the inverse label frequency: *Love* 3.87, *Joy* 1.22, *Surprise* 20.72, *Anger* 4.20, *Sadness* 3.03, and *Fear* 65.02. This up-weights the gradient contribution of the rare *Fear* and *Surprise* classes so that they are not dominated by the frequent labels.

The learning rate was annealed with a ReduceLROnPlateau scheduler (factor 0.5, patience 2, minimum 10^{-6}) monitoring the validation loss, gradients were clipped to a maximum norm of 1.0, and training was stopped early if the validation macro-F1 did not improve for five consecutive epochs, restoring the best model state. The quantum circuit parameters and the classical weights were optimized jointly in a single backward pass.

4.5 Evaluation Metrics

We evaluate all configurations with three complementary schemes. *Scheme 1* is the per-label evaluation based on scikit-learn: the macro, micro, and weighted F1-scores together with the Hamming loss. For a test set of N samples with K labels, let $y_i \in \{0, 1\}^K$ and $\hat{y}_i \in \{0, 1\}^K$ denote the gold and predicted label vectors. The *Hamming loss* is

$$\text{HammingLoss} = \frac{1}{NK} \sum_{i=1}^N \sum_{k=1}^K \mathbb{1}[\hat{y}_{ik} \neq y_{ik}], \quad (4)$$

and the macro F1-score is the unweighted mean of the per-label F1-scores.

Scheme 2 is the *Exact Match Ratio* (EMR), which scores a sample as correct only if its entire predicted label vector matches the gold vector:

$$\text{EMR} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}_i = y_i]. \quad (5)$$

Because there is no partial credit, precision, recall, and F1 coincide with the EMR for this strict measure.

Scheme 3 is an instance-level *partial-match* score that gives partial credit when the predicted set of positive labels overlaps the gold set without being identical. For each sample, the active label sets $G_i = \{k : y_{ik} = 1\}$ and $P_i = \{k : \hat{y}_{ik} = 1\}$ are intersected, and per-instance precision, recall, and F1 are computed as

$$P_i = \frac{|G_i \cap P_i|}{|P_i|}, \quad R_i = \frac{|G_i \cap P_i|}{|G_i|}, \quad F1_i = \frac{2P_i R_i}{P_i + R_i}, \quad (6)$$

with the edge cases that both-empty sets yield a perfect score of 1 and a one-empty set yields 0. The reported partial-match precision, recall, and F1 are the macro averages of P_i , R_i , and $F1_i$ over the test set.

5 Experiments and Results

5.1 Experimental Setup

All experiments were performed using the PyTorch and PennyLane libraries. The transformer encoders ran on a pair of NVIDIA T4 GPUs, while the variational quantum circuits were executed on a CPU using PennyLane’s `default.qubit` simulator, with circuit outputs moved

back to the GPU after each forward pass. A random seed of 42 was fixed across Python, NumPy, and PyTorch (including CUDA) for reproducibility. For evaluation, we report macro, micro, and weighted F1-scores together with the Hamming loss, and additionally the Exact Match Ratio (EMR) and the instance-level partial-match precision, recall, and F1-score (Section 4.5). All models were evaluated on the same held-out test partition of 2,272 comments.

5.2 Results

The performance of the classical baseline and the quantum variants is reported in Table 2. For the stronger IndicBERTv2 encoder, the classical baseline achieves a macro F1-score of 0.5333 and an Exact Match Ratio of 0.4252. The quantum configurations do not surpass this macro F1-score, with the best value of 0.5165 obtained by Custom 2 at eight qubits; however, Custom 2 at eight qubits also achieves the highest Exact Match Ratio of 0.4577, indicating that the quantum branch improves the joint subset-prediction accuracy even when the per-label average is comparable.

For the weaker BanglaBERT encoder, the quantum variants provide substantial gains over the classical baseline. The classical model attains only a macro F1-score of 0.2101 and an Exact Match Ratio of 0.0000, whereas Custom 2 reaches a macro F1-score of 0.4850 and an Exact Match Ratio of 0.4630. The trend in Fig. 4 shows that increasing the qubit count is most beneficial when combined with an expressive encoding and entanglement strategy; increasing the register size alone does not guarantee uniform improvement.

Table 2: Performance comparison of IndicBERTv2 [3] and BanglaBERT [1] under different qubit and VQC configurations.

Framework	Qubit	VQC	Macro			Partial Match			Exact Match Ratio
			Precision	Recall	F1-score	Precision	Recall	F1-score	
IndicBERTv2	0	N/A (Classical)	0.4917	0.5900	0.5333	0.6136	0.6916	0.6268	0.4252
	2	SEL	0.2861	0.6220	0.3741	0.3564	0.7606	0.4739	0.0238
		Custom 1	0.4493	0.6406	0.4869	0.6039	0.6820	0.6164	0.4230
		Custom 2	0.3758	0.5720	0.4443	0.4601	0.7225	0.5386	0.1527
	4	SEL	0.4032	0.5796	0.4572	0.4825	0.6704	0.5387	0.2148
		Custom 1	0.4616	0.5496	0.4922	0.5996	0.6845	0.6141	0.4155
		Custom 2	0.3717	0.6115	0.4523	0.4062	0.7476	0.5156	0.0519
	8	SEL	0.3528	0.6212	0.4434	0.4082	0.7559	0.5189	0.0497
		Custom 1	0.4642	0.5521	0.4982	0.6030	0.6814	0.6144	0.4265
		Custom 2	0.5095	0.5301	0.5165	0.6266	0.6685	0.6255	0.4577
BanglaBERT	0	N/A (Classical)	0.1358	0.5000	0.2101	0.2716	0.7144	0.3870	0.0000
	2	SEL	0.2916	0.6392	0.3825	0.3637	0.7695	0.4822	0.0264
		Custom 1	0.3946	0.6033	0.4486	0.4521	0.7111	0.5320	0.1505
		Custom 2	0.1760	0.6667	0.2749	0.2640	0.9198	0.4039	0.0000
	4	SEL	0.4369	0.5713	0.4791	0.5407	0.6902	0.5796	0.3178
		Custom 1	0.4419	0.5136	0.4653	0.5710	0.6591	0.5858	0.3891
		Custom 2	0.4622	0.5174	0.4850	0.5954	0.6652	0.6062	0.4071
	8	SEL	0.3521	0.5994	0.4347	0.3974	0.7475	0.5093	0.0387
		Custom 1	0.4501	0.5001	0.4715	0.5809	0.6599	0.5935	0.3895
		Custom 2	0.4915	0.4780	0.4798	0.6175	0.6371	0.6085	0.4630

Across all configurations in Table 2, the quantum branch consistently improves the Exact Match Ratio over the corresponding classical baseline whenever the encoding and entanglement structure are well matched to the encoder, with the largest joint-subset gains observed at the eight-qubit register. The full implications of these trends are examined in the Discussion.

6 Discussion

The results demonstrate that both the pretrained encoder and the quantum-circuit design strongly influence classification performance. IndicBERTv2 provides the strongest classical representation, whereas the quantum variants produce particularly large gains for the weaker BanglaBERT baseline. As the qubit count increases, Custom 1 shows relatively consistent improvement, while Custom 2 achieves the highest exact-match ratios for both encoders, reaching 0.4577 with IndicBERTv2 and 0.4630 with BanglaBERT, and also records the largest single gain over the classical baseline (EMR 0.4630 versus 0.0000 for BanglaBERT). The trend in Fig. 4 further indicates that additional qubits are most beneficial when paired with an expressive encoding and entanglement strategy; increasing circuit size alone does not guarantee uniform improvement across all configurations.

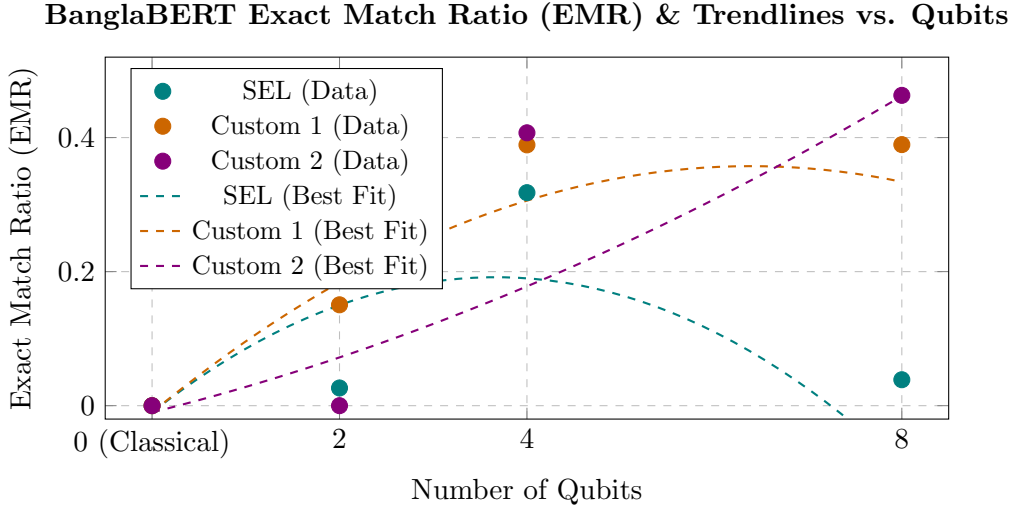


Figure 4: BanglaBERT Exact Match Ratio (EMR) across qubit counts for SEL, Custom 1, and Custom 2, together with their fitted trend lines.

6.1 Design Variations

Further changes were developed independently to improve the two custom circuits. Custom 2 was handled by Tathagata Mitra, while Custom 1 was handled by Biswanath Ranjit.

Custom 1 improvements. The following changes were introduced to increase the expressivity, training stability, and information-preservation capacity of Custom 1 while retaining its four-qubit register size:

- **Deep feature projection network:** IndicBERT’s 768 contextual features are compressed to four VQC inputs through a multi-stage linear architecture, $768 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 4$, using GELU activations and a dropout rate of 0.3 to minimise information loss and reduce overfitting.
- **Enhanced variational quantum circuit:** The circuit depth is increased to two variational layers with independent trainable parameters. Inputs are encoded through `Y-axis AngleEmbedding`, rotations are bounded using $\tanh(\cdot)$ input normalisation, and cross-entanglement connections ($0 \rightarrow 2$) and ($1 \rightarrow 3$) are added alongside the standard ring entanglement.

- **Post-quantum fusion layer:** A learnable linear layer, $4 \rightarrow 128$, is appended after the VQC to project the four quantum expectation values into a higher-dimensional classical representation for classification.
- **Differential optimisation strategy:** Component-specific learning rates are used for IndicBERT (2×10^{-5}), the classifier (1×10^{-4}), and the quantum VQC (1×10^{-3}). These are combined with a weight decay of 0.01, `ReduceLROnPlateau` scheduling, and early stopping with a patience of five epochs.
- **Multi-label class-imbalance handling:** `BCEWithLogitsLoss` is configured with label-frequency-based `pos_weight` values to improve the detection of underrepresented emotions such as Fear and Surprise.
- **Validation-threshold optimisation:** The static decision threshold of 0.5 is replaced with per-emotion optimal thresholds, ranging from 0.26 for Surprise to 0.82 for Sadness, to maximise per-class F1-scores.
- **Performance benchmark gains:** On the four-qubit configuration, these changes improve performance across the primary metrics and raise the Exact Match Ratio from 0.4155 to 0.4674.

Custom 2 improvements. The following changes were introduced to increase the expressivity and information capacity of Custom 2 without increasing its four-qubit register size:

- **Measurement strategy:** The baseline circuit measured only the single-qubit Pauli- Z and Pauli- X expectation values, yielding eight output features. The measurement set is expanded to include the Pauli- Y expectation on every qubit and the four two-qubit ring correlators $\langle Z_i Z_{i+1} \rangle$. This produces 16 measurement outputs and transfers richer information from the quantum subsystem to the classical readout without increasing the number of qubits.
- **Data encoding:** The input encoding is extended from two-axis rotations (R_X, R_Y) to three-axis rotations (R_X, R_Y, R_Z) on each qubit, increasing the number of classical features embedded into the quantum state from eight to twelve. A learned phase rotation mixes the first feature of each pair before encoding, while the encoding scale remains trainable.
- **Data re-uploading:** The number of data re-upload repetitions is increased from one to two per variational layer. This allows each encoding step to interact with the trainable unitary more frequently and increases the effective nonlinearity of the quantum feature map.
- **Variational ansatz:** The entangling pattern is made layer-dependent. Even-indexed layers use a nearest-neighbour cycle of controlled- Z gates, whereas odd-indexed layers use controlled-NOT gates between next-nearest neighbours. Alternating the entangling topology reduces redundancy between successive unitary blocks and improves circuit expressibility.
- **Layer count:** The number of variational layers is increased from one to three. Each layer now uses its own dedicated rotation parameters, rather than repeatedly using a shared weight slice, ensuring that the additional circuit depth provides additional trainable capacity.
- **Performance benchmark gains:** Together, these changes raise the Exact Match Ratio for Custom 2 from 0.4055 to 0.4804.

7 Conclusion

This paper investigated hybrid quantum-classical architectures for multi-label emotion classification of Bengali text, a low-resource setting in which conventional baselines plateau in the 50–60% range. Two custom variational quantum circuits were designed and evaluated under an identical outer framework: the Custom 1 cross-coupling VQC, which combines a simple R_Y angle encoding with two entangling layers using distinct nearest-neighbour and cross-coupling CNOT topologies; and the Custom 2 re-uploading VQC, which combines R_X/R_Y encoding with learnable per-qubit phase and scale parameters, a rotating ansatz, and input re-injection between variational layers.

Three main conclusions follow from the experimental study. First, the learned representation space depends strongly on the selected classical encoder and quantum embedding. The IndicBERTv2 classical baseline achieved a macro F1-score of 0.5333, compared with 0.2101 for BanglaBERT. The strongest quantum configurations reached macro F1-scores of 0.5165 for IndicBERTv2 and 0.4850 for BanglaBERT, while the highest exact-match ratios were 0.4577 and 0.4630, respectively, all achieved by the re-uploading Custom 2 design. These results show that the quality of the pretrained representation remains important, although the quantum configurations substantially improve the weaker BanglaBERT baseline.

Second, scaling behaviour is architecture- and encoder-dependent. The Custom 1 macro F1-score improves consistently as the register grows from two to eight qubits for both IndicBERTv2 (0.4869 to 0.4982) and BanglaBERT (0.4486 to 0.4715). Custom 2 exhibits a larger but less uniform gain, with its strongest performance appearing at eight qubits for IndicBERTv2 and four qubits for BanglaBERT; at the eight-qubit register its exact-match ratio exceeds that of Custom 1 for both encoders (0.4577 versus 0.4265 and 0.4630 versus 0.3895). This suggests that additional Hilbert-space capacity is useful only when the encoding, entanglement pattern, and classical representation are jointly compatible, and that the re-uploading scheme exploits this capacity most effectively.

Third, simulator-backend selection does not alter the reported analytic scores. The PennyLane `default.qubit` simulator is used for all reported results, and switching between mathematically equivalent simulator backends reduces wall-clock training time from several hours to tens of minutes without changing the model outputs. The experiments also indicate that a four-qubit circuit trained with adjoint differentiation is preferable to backpropagation on PennyLane’s accelerated devices. AdamW optimization, plateau-based learning-rate reduction, and early stopping provide a stable training protocol for the hybrid model on the strongly imbalanced EmoNoBa corpus.

8 Limitations

The principal limitation is that the quantum branch introduces relatively few trainable parameters, so its benefit depends heavily on the quality of the compressed classical features. For very small registers, the quantum contribution can be dominated by the projection and post-processing layers. In addition, all experiments were conducted on quantum simulators rather than on physical hardware. Future work should extend the comparison to larger registers with controlled re-uploading depth, combine cross-coupling entanglement with data re-uploading in a single ansatz, incorporate multi-observable and correlated readouts, and evaluate the strongest architectures under finite-shot and noisy simulation before deployment on quantum hardware.

A further limitation concerns the evaluation protocol itself. The Exact Match Ratio is an intentionally strict criterion that assigns a score of zero to any sample for which the predicted label set differs from the gold label set in even a single emotion, and the reported partial-match scores are macro-averaged over instances without instance-level confidence weighting. Both measures therefore depend on the choice of the decision threshold, and the per-label thresholds

used here were tuned on the validation partition, which may not transfer perfectly to other data distributions. The dataset also spans a single language and a limited set of social-media domains, so the conclusions may not generalise to other low-resource languages, longer texts, or formal registers without retraining.

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